Privacy-Enhanced Synthetic Data Generation for Autonomous Vehicle Training

R26-CS-010

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Comments for the Progress Presentation 1

Topic	Privacy-Enhanced Synthetic Data Generation for Autonomous Vehicle Training			
Project ID	R26-CS-010			
Specialization	CS			
Student ID	IT22309556	IT22541284	IT22110220	IT22066916
PanelName	CS-P02			
Detailed Comments	You shold improve your presentation skills. (Make sure to keep eye contact with the pannel members/ Poor in time management) They are using a standard format for encryption, is it better to use new algorithm or concept to prove that system is not vulnarable Presentation slides are not in standard, you should use proper style for when prepairing the slides.	Presentation slides are not in standard, you should use proper style for when prepairing the slides. They are using a standard format for encryption, is it better to use new algorithm or concept to prove that system is not vulnarable	Presentation slides are not in standard, you should use proper style for when prepairing the slides. They are using a standard format for encryption, is it better to use new algorithm or concept to prove that system is not vulnarable	Presentation slides are not in standard, you should use proper style for when prepairing the slides. They are using a standard format for encryption, is it better to use new algorithm or concept to prove that system is not vulnarable

THE PROBLEM





IT22309556 | S. A. S. D. PRIYADARSHI

Individual Component: **Privacy Integration**

Novelty:

- Privacy enforced during data generation rather than post-processing.

PROBLEM , GAP & SOLUTION

Enforcing and validating differential privacy in the AV data pipeline

THE PROBLEM

Simulators are configured from real driving data

Privacy protects outputs, not that configuration

THE GAP

No AV platform offers a formal guarantee

DP-GAN tested on MNIST only, not AV data

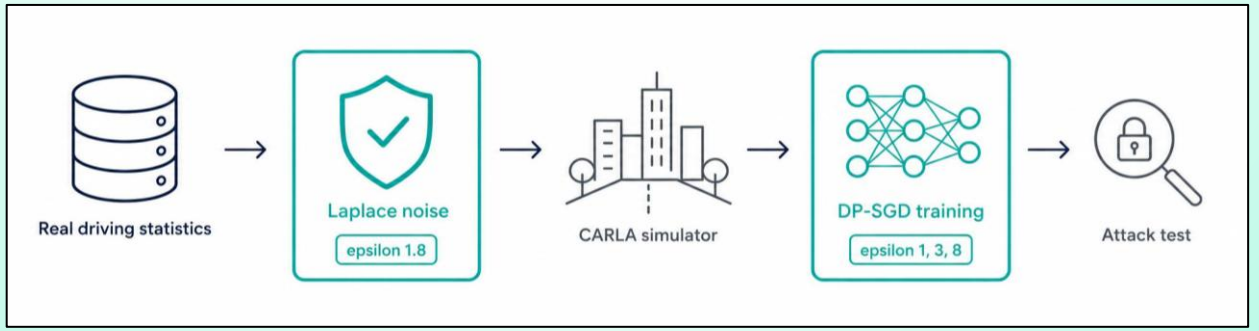
Nobody attacks it to check

MY SOLUTION

Laplace noise before the simulator

DP-SGD during training, budget auto-enforced

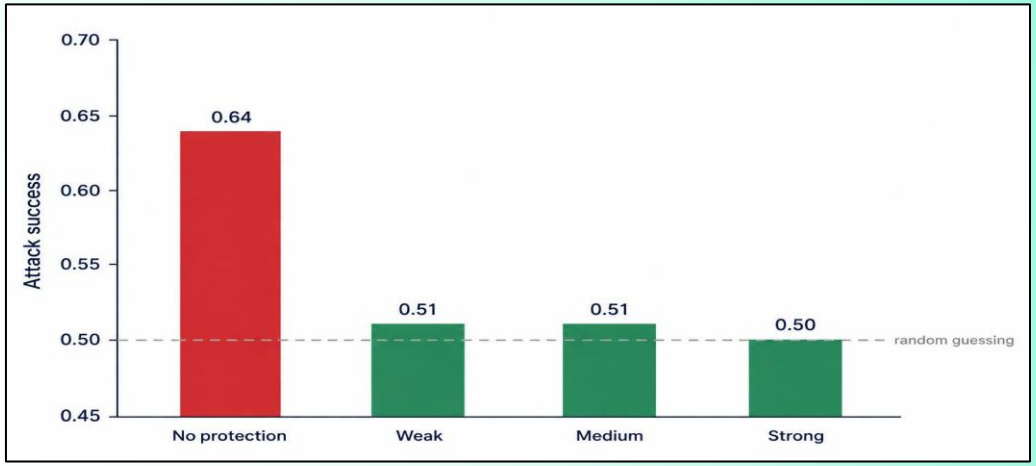
Attack it to prove it holds



PRIVACY INTEGRATION

PROGRESS AND RESULTS

PROTECTION	ε	ACCURACY	ATTACK
None	—	0.83	64%
Weak	8.0	0.67	51%
Medium	3.0	0.66	51%
Strong	1.0	0.62	50%



COMPLETED

Full pipeline implemented and tested
3 independent runs x 4 configurations
Statistically significant ($p < 0.01$)
Reproducibility verified, certificate issued

FUTURE WORK

More runs to separate the strengths
Combine with utility results (IT22110220)
Independent verification (IT22066916)
Final compliance certification

The leak was removed. It cost 0.17 accuracy. Both measured, not assumed.



IT22541284 | YUBITHA S.

Individual Component: **Secured Dataset Generation**

Novelty:

- Generation of **privacy-aware synthetic multi-sensor datasets directly from the simulator environment**, enabling realistic AV training data while preserving privacy constraints.

SECURED DATASET GENERATION

THE PROBLEM

Autonomous-vehicle models need large volumes of real-world driving data

That data exposes sensitive details like pedestrian identities, plates, GPS traces

Sharing it across teams creates privacy, ethical, and cybersecurity risk

THE GAP

Existing synthetic-data systems optimize mainly for realism and performance

Privacy and security are bolted on afterward, not designed in

Result: datasets remain open to tampering, leaks, and unauthorized access

MY SOLUTION

CARLA-based generation across five towns with multi-modal sensor capture

Automated ground-truth labeling for detection and segmentation models

Security built into the pipeline:
SHA-256 → AES-256-GCM → RBAC

Secure Handover and Independent Validation

How the pipeline protects data end to end and proof it holds up

Secure Dataset Handover

- Encrypted hash verified batches let researchers confirm integrity before training
- Synthetic data removes exposure to real identities plates and locations
- SHA 256 fingerprinting and AES 256 encryption block tampering and unauthorized reads
- RBAC gives each role simulation privacy ML only the access it needs

Independent Security Validation

- 6 test reproducible cryptographic audit claims backed by data not opinion
- Perfect round trip recovery unique IVs and ideal avalanche diffusion confirmed
- GCM's built in MAC blocks tampered ciphertext 6,093 real files scanned 0 vulnerabilities
- Result 5 of 5 properties verified upgraded to authenticated AES 256 GCM

Future Works

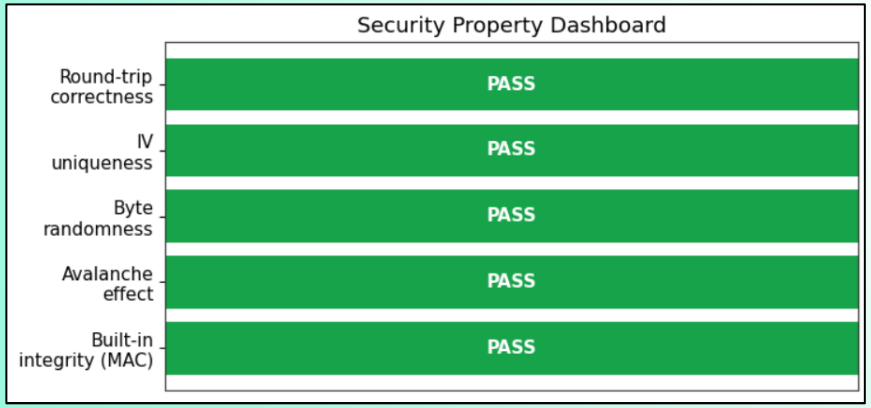
- Using AI autonomous system to collect real world datasets.

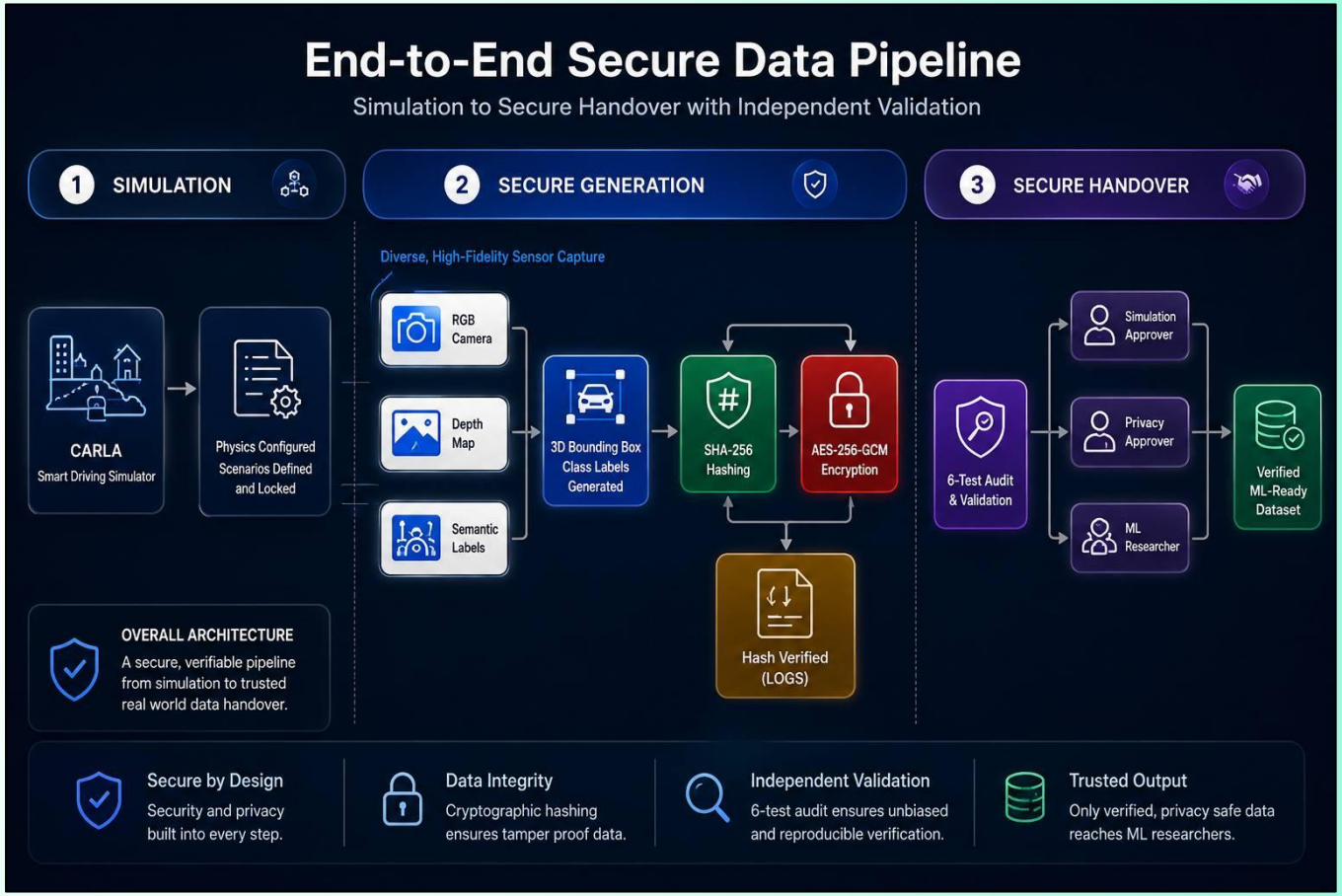
PP1
Comment:

Presentation slides are not in standard, you should use proper style for when prepairing the slides.

They are using a standard format for encryption, is it better to use new algorithm or concept to prove that system is not vulnarable

How I Address that:







IT22110220 | M.K.S.R. APEKSHA

Individual Component: **Utility Evaluation**

Novelty:

- Framework generating privacy–enhanced synthetic datasets evaluating AV perception privacy–utility trade–off.

Utility Evaluation - IT22110220

PROBLEM , GAP & SOLUTION

THE PROBLEM

AV models need millions of annotated real road images.

Real footage exposes PII (faces, plates) -> GDPR / PDPA risk.

CARLA synthetic data fix reintroduces privacy risk via calibration.

THE GAP

Privacy alone doesn't guarantee usability. Pure noise is private but worthless.

"Private" claimed, but usability never quantitatively proven.

Missing: an independent, evidence-based answer to "Is synthetic data good enough vs. real data?"

MY SOLUTION

Stage	Solution
Decryption + integrity check	40,000 / 40,000 files , 0 failed, SHA 256 verified
Exploratory data analysis	Class imbalance (~4.5:1);86.4% pedestrians <32x32px
Poisoning detection (security)	15,084 objects scanned, 0 anomalies, Pass proven via deliberate poisoning test
YOLO format conversion	10,000 / 10,000 frames, reproducible 70/15/15 split
YOLOv8 training (synthetic)	CPU trained , best.pt / last.pt saved
mAP measurement (test split)	mAP50-95 = 0.2273, confirms class imbalance effect
Size stratified mAP analysis	Small object size:– not model failure drives weak pedestrian mAP
KITTI real data baseline	7,481 pairs converted / split for synthetic vs. real comparison

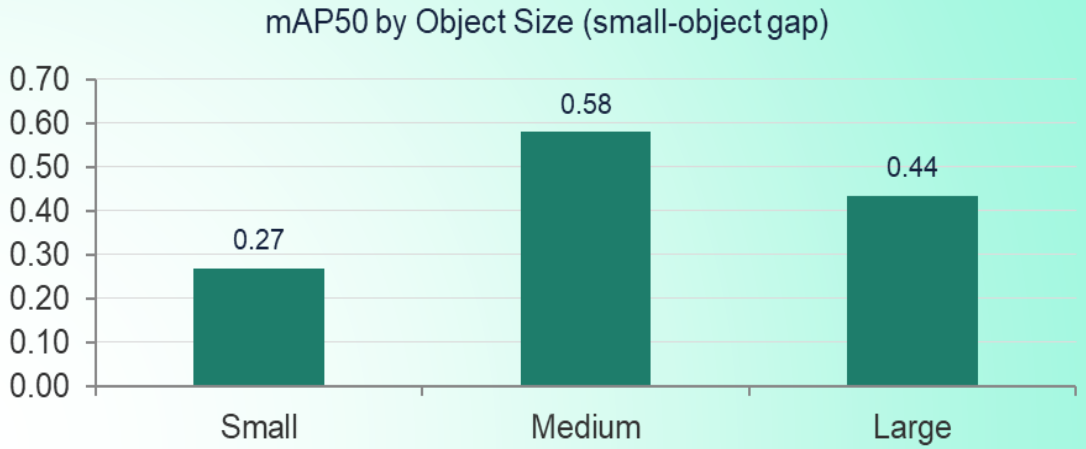
Result: Verified, quantitative foundation proving synthetic data can train a functional AV perception model and sim to real comparison (KITTI baseline) in progress for the heading finding.

UTILITY EVALUATION

PROGRESS & RESULTS

✓ COMPLETED

Dataset decrypted & integrity verified (40,000 files, SHA-256)
Class imbalance & small object analysis completed
Poisoning detection audit: 15,084 boxes, 0 anomalies that are PASS
YOLO dataset built (70/15/15 split, seed=42)
YOLOv8n trained (CPU, 4 epochs) & official mAP measured
KITTI real data baseline: converted & Started to trained



CLASS	AP50	AP50-95	NOTE
All (overall)	0.3655	0.2273	Primary metric
Car	0.6076	0.3981	Best - largest class
Pedestrian	0.1793	0.0955	Weakest - small objects
Cyclist	0.3096	0.1882	Mid-range

PP1 Comment:

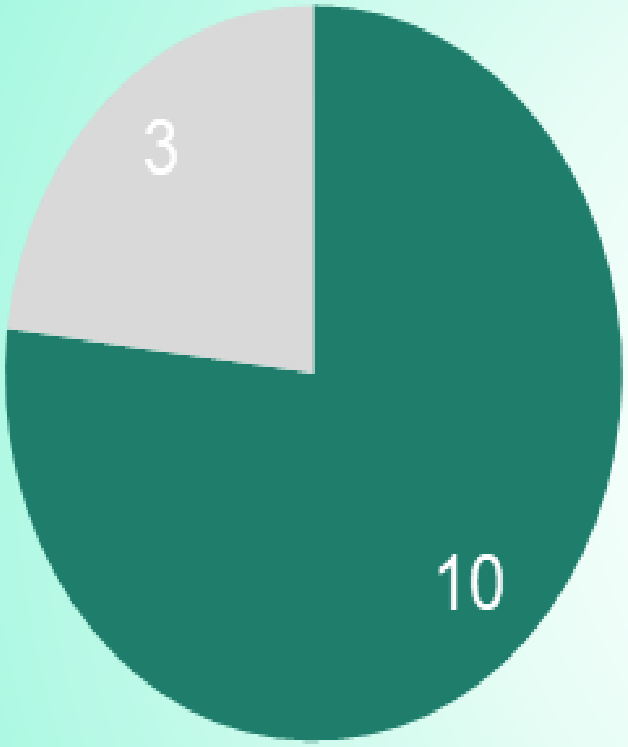
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Cybersecurity control - data poisoning detector: 0 anomalies in 15,084 boxes audited (PASS). Deliberate poisoning test: 3 of 3 injected anomalies correctly flagged.

UTILITY EVALUATION

FUTURE WORK



EXPERIMENT	TRAIN	TEST	STATUS
1. Baseline	Synthetic	Synthetic	Done (Part 8-9)
2. Sim-to-real	Synthetic	KITTI	Pending - key number
3. Upper bound	KITTI	KITTI	Pending
4. Reverse gap	KITTI	Synthetic	Pending

✓ **PLANNED (Part 11-13)**

Run experiments 2-4 using trained synthetic (Part 8) & KITTI (Part 10) models
Compute sim-to-real mAP gap as % of the real-to-real baseline (headline result)
Compute FID between synthetic and KITTI test images (n=1,000 per set)
Report per class honesty: car vs pedestrian/cyclist AP gap, explained by imbalance & object size
State the KITTI aspect ratio confounder explicitly in the write up
Compile final results table & connect findings to DP-SGD results (IT22309556)

Confounder to report: KITTI images (1242x375) vs CARLA images (800x600) differ in aspect ratio and field of view - part of the measured sim-to-real gap comes from this, not privacy alone.



IT22066916 | R.D.P.T. WIJESINGHE

Individual Component: **Risk and Compliance**

Novelty:

- Introduces an **attack-based privacy evaluation method** for synthetic AV data pipelines.

PROBLEM , GAP & SOLUTION

Enforcing and validating differential privacy in the AV data pipeline

THE PROBLEM

AV synthetic data pipelines claim privacy but never prove it.

Real driving data exposes faces, plates and GPS traces.

GDPR Article 25 requires proof - not assumption.

THE GAP

No independent attack testing framework exists for AV pipelines.

Privacy controls added after generation - never tested against real attacks.

Compliance claims cannot be defended without attack evidence.

MY SOLUTION

Built independent privacy attack framework - 6 attacks across the pipeline.

Certificate verification, budget audit, reconstruction, MIA replication.

Every attack produces JSON + PDF evidence mapped to GDPR articles

- ✓ GDPR & NIST compliance baseline documented.
- ✓ 5 privacy attack scripts built and executed.
- ✓ 5 attack reports generated.
- ✓ All results mapped to GDPR articles and NIST framework.

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Results & Maturity

Claims Supported by data – limitation acknowledged

Attack	Result	Safe Range	Status
Attack 1 -Certificate Verification	4/4 PASS, Tamper Detected	All sigs must match	PASS
Attack 2 – Privacy Budget Audit	ε matched across 2200 step	Reported = Logged ±0.0001	PASS
Attack 3 -Reconstruction Attack	Ratio 0.94	Ratio 0.5 – 1.5 = safe	PASS
Attack 4 - Independent MIA	DP AUC 0.50–0.53	AUC < 0.55 = safe	PASS
Attack 5 - Metadata Leakage	0 findings / 30,000 images	0 high severity findings	PASS
Attack 6 - Re-Identification	Pending...	ReID rate < 5% = safe	Pending...

2. Limitations & Next Steps.

- Attacks 6 pending - waiting for IT22110220's Outputs.
- Fine-tune all attack scripts for final evaluation.
- Produce final compliance report PDF.

Commercialization

Target buyers: AV/ADAS companies, university/industry research labs, and government transport-safety bodies needing training data without real-world privacy liability

Product form: Licensed "Secure Synthetic Dataset-as-a-Service" : subscription access to encrypted, hash-verified, calibrated CARLA datasets by town/scenario type

Value proposition: Removes the cost and legal risk of collecting and anonymizing real driving footage: GDPR-type compliance built in by design, not bolted on

Differentiator vs. competitors: Security-by-design pipeline (SHA-256 + AES-256-GCM + RBAC) with an independently auditable validation report , not just a data generator

Growth path: Start with bounding-box + segmentation-labeled datasets for AV perception teams -> expand to a plug-in CARLA module other researchers can license directly



Thank You!